Executive Summary: Analyzing the Impact of Precipitation on Flight Departure Delays at JFK

A Data-Driven Operational Evaluation Using the JFK International Airport Dataset

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1. Purpose

Management requested an operational analysis to evaluate how rainy weather conditions relate to flight departure delays at John F. Kennedy International Airport (JFK). Understanding the correlation between precipitation and scheduling disruptions is critical for optimizing ground handling procedures, adjusting turnaround times, and improving passenger communication during inclement weather.

2. Key Finding

By evaluating flight departure delays against hourly weather logs, the analysis reveals that flights departing during periods with recorded precipitation experience substantially higher average departure delays and a much higher percentage of significant delays (exceeding 15 minutes) compared to clear-weather conditions. Specifically, flights during clear weather averaged 12.63 minutes of departure delay, with 21.53% delayed more than 15 minutes, while flights during rain averaged 113.11 minutes of delay, with 88.89% delayed more than 15 minutes. It is worth noting that only 18 flights in the dataset were recorded as departing during rain, compared to 326,975 during clear weather, so this rain-condition average should be read as directional rather than statistically robust. The SQL query below extracts, categorizes, and aggregates this operational impact.

SQL Query Used

SQL

```
WITH flight_weather_delay AS (
   SELECT
      f.carrier,
      f.dep_delay,
      CASE WHEN w.precip > 0 THEN 'Raining' ELSE 'Clear' END AS rain_condition
   FROM jfk.flights f
   JOIN jfk.weather w
     ON w.origin = f.origin
    AND w.year = f.year
    AND w.month = f.month
    AND w.day = f.day
    AND CAST(w.hour AS INTEGER) = CAST(f.hour AS INTEGER)
```

```
    WHERE f.dep_delay IS NOT NULL
      AND w.precip IS NOT NULL
)
SELECT
    rain_condition,
    COUNT(*) AS total_flights,
    ROUND(CAST(AVG(dep_delay) AS NUMERIC), 2) AS avg_departure_delay_mins,
    ROUND(CAST(100.0 * SUM(CASE WHEN dep_delay > 15 THEN 1 ELSE 0 END) / COUNT(*)
AS NUMERIC), 2) AS pct_delayed_over_15min
FROM flight_weather_delay
GROUP BY rain_condition;
```

weather_condition	total_flights	average_departure_delay	percent_delayed
Clear	326975	12.63	21.53
Raining	18	113.11	88.89

3. Defensible Position & Evaluation

Three Reasons Why This Is a Good Metric

1. **Directly Links Environmental Factors to Operations:** Categorizing flights by precipitation status (precip > 0) creates a clear, binary baseline that allows management to directly observe the operational drag caused by weather.
2. **Utilizes Standardized Thresholds (15-Minute Rule):** Tracking both average delay minutes and the percentage of flights delayed beyond 15 minutes aligns with standard aviation industry benchmarks (such as Department of Transportation on-time metrics).
3. **Normalizes Large Volume Datasets:** Grouping by condition across thousands of flights prevents isolated extreme weather anomalies from distorting the overall operational picture.

Three Limitations of the Metric

1. **Binary Rain Classification Lacks Intensity Details:** A light drizzle is treated identically to a severe downpour or thunderstorm, even though heavy convective storms cause much more severe ground stops.
2. **Hourly Weather Granularity:** Weather data is recorded on an hourly basis, whereas flight departures occur at specific minutes within that hour, creating a slight temporal mismatch between the recorded precipitation and pushback time.
3. **Cascading Network Delays:** A flight departing late in the rain may be delayed not because of local rain at JFK, but because its incoming aircraft was delayed upstream at another airport experiencing bad weather.

What Additional Data Would Improve This Metric?

To enhance this analysis, we would benefit from incorporating rainfall intensity rates (inches per hour), radar or lightning strike data to capture severe thunderstorms, runway-specific de-icing and visibility metrics, and upstream aircraft routing logs to isolate local weather delays from network-wide schedule propagation.

4. Conclusion

The data suggests a strong association between precipitation and increased departure delays at JFK International Airport, though the very small number of rain-affected flights in this dataset means that association should be confirmed with a larger sample before being treated as a firm conclusion. While this query provides a strong foundational baseline for weather-impact analysis, pairing it with storm intensity and aircraft routing data in future studies will allow management to build even more resilient scheduling models.